Physics-Based Gun Barrel Wear Monitoring and Life Prediction System

Field of the Invention

The current invention has to do with the defense structural health monitoring systems, and specifically physics-based thermo-mechanical model in real time gun barrel wear monitoring, and prediction of service life. The invention is devoted to assessing the structural degradation of the medium-caliber gun barrels i.e., 30 mm autocannon systems that are subjected to extreme firing conditions with high internal pressure, high temperature increase, and cyclic loading.

Gun barrels experience high thermo-mechanical stresses during firing operations that comprise radial pressure stress, hoop stress, longitudinal stress, and thermal expansion. The result of these stresses is material fatigue, formation of micro-cracks, bore deformation and erosion of the barrel surface. With time, such degradation has the effect of decreasing structural integrity, firing precision, and the potential cause of catastrophic failure, unless it is well checked.

The current invention offers a physics-based analytical monitoring scheme combining temperature detection, strain observation, vibration monitoring, and underlying computational modelling to gauge the wear progression and forecast the rest of the service life of the barrel. The system uses the known engineering concepts such as thick-cylinder stress analysis, thermal stress analysis, calculation of strain energy, and modeling of cumulative fatigue damage to calculate structural degradation in the operational firing cycles.

The proposed system facilitates continuous structural assessment and predictive maintenance, in contrast to the traditional method of maintenance, which is based on infrequent replacement of its parts on a fixed firing cycle or manual examination. The invention is based on real-time analysis of thermo-mechanical behavior of the barrel, which makes it possible to estimate the wear percentage, status of structural health and the rest of the safe firing cycles.

Defense weapon systems (principally naval autocannons, ground-based systems of medium caliber weapons and other structures of high pressure) are also of interest to the invention, wherein the stability of the structural integrity is mandatory to the operational safety, optimization of maintenance, and enhanced life service utilization.

Background of the Invention

Medium caliber armament systems like 30 mm auto guns have gun barrels that are subjected to very harsh thermo-mechanical environments. High internal pressures, heat, cyclic loading and repeated shock waves caused by the discharge of projectiles are experienced on the barrel during firing. Such severe environments slowly lead to changes of structure on the barrel material, radial expansion, build-up of thermal stress, micro-crack formation, fatigue damage and bore surface erosion. When these degradations occur in extended firing cycles, they may cause severe damage to the structural integrity and the operational life of the gun barrel.

Conventional gun barrel maintenance principles are mainly based on predetermined firing-cycle replacement intervals or intermittent manual checking systems like bore scoping, dimensional checks and assessment with naked eyes. Even though these methods aid in keeping things safe in the operations of the barrel, they tend to be inefficient in that they fail to give a true picture of the actual structural state of the barrel on the operation. The actual fatigue life is often underutilized and the costs of its maintenance are higher due to premature replacement of barrels with very conservative safety margins. On the other hand, lack of proper monitoring can also lead to chances of structural failure that occurs unpredictably in cases of critical operations.

The majority of the current work and industrial solutions aim at enhancing barrel life by using superior materials, coating methods, chrome lining, autofrettage, which increase mechanical and wear resistance. Although the improvements will increase the barrel life, it does not represent a stable form of continuing structural monitoring, as well as real-time life prediction. Also, a wide range of surveillance strategies do not include the thermo-mechanical analysis of stresses, strain-energy analysis, and fatigue damage accretion, whose analysis is incorporated into a single analytical platform.

Hence, there is a requirement of an effective monitoring system that can continuously assess the structure of gun barrels in actual shooting. This requirement is addressed by the current invention, which presents an analytical system based on physics that will combine temperature measurements, strain measurements, vibration measurements, and thermo-mechanical modeling to approximate wear behavior and forecasting of the remaining service life. This will make it possible to have predictive maintenance, enhance safety, and improve the efficient exploitation of gun barrel service life in defense functions.

Summary of the Invention

The current invention refers to a Physics-Based Gun Barrel Wear Monitoring and Life Prediction System that is meant to measure the degradation of medium caliber gun barrels that are subjected to a cyclic high-pressure firing regime. The firing of gun barrels is characterized by intense thermo-mechanical loads such as high internal pressure, high temperature increase, and cyclic shock loading which over time cause material fatigue, bore deformation as well as structural wear. Careful control over these conditions is necessary to provide safety of operation and correct use of barrel service life.

The invention proposes a physics-based analytic tool that continually measures the health of the barrels through thermo-mechanical concepts instead of just using set replacement cycles of the firing cycles or manual inspection techniques. The system incorporates a temperature sensor, strain gauges, and vibration measurements units to record real-time performance of the barrel when shooting is taking place. The embedded microcontroller processes these sensor inputs, then does stress analysis and fatigue assessment of the sensor inputs based on standard mechanical engineering models.

The analytical engine uses the equations of radial, hoop and longitudinal stresses which are developed under internal firing pressure using thick-cylinder stress equations. Elastic expansion relations are used to assess the thermal stress by depending on temperature change in the barrel structure. Besides that, strain energy density and cumulative fatigue damage are also determined in order to estimate the wear progression during repeated firing cycles.

According to the calculated results, the system produces a Health Index which is the structural condition of the barrel and predicts the rest of safe firing cycles. The monitoring structure also offers the classification results like SAFE, WARNING and CRITICAL where the operators and the maintenance staff can take the necessary preventative measures.

The invention has offered an effective and dependable predictive maintenance solution to the defense weapon systems in the form of a transparent and real-time sensor readings by combining thermo-mechanical modeling and real-time sensor measurements. The system elevates the operational safety, minimal frequency of replacing the barrels ahead of time, and the overall efficiency of management of barrel life in the military operations.

Objects of the Invention

The main goal of the current invention is to come up with a physics-based gun barrel wear sensor and life forecast platform that can possibly assess structural degradation in medium-caliber gun barrels that are fired cyclically at high pressure.The other aim of the invention is to offer a real time monitoring structure that constantly assesses the thermo-mechanical stress like radial stress, hoop stress, longitudinal stress and thermal stress, which occurs during firing operations.

Another purpose of the invention is to combine sensing temperature with strain gauging and vibrations to gather data about operations of the gun barrel and provide the opportunity to analyze the actions of the structure correctly during repeated firing cycles.The other aim of the invention is to achieve the analytical stress modeling using thick-cylinder theory and relations on thermal expansion to establish the deformation, stress distribution, and strain energy build-up in the material of the barrel.

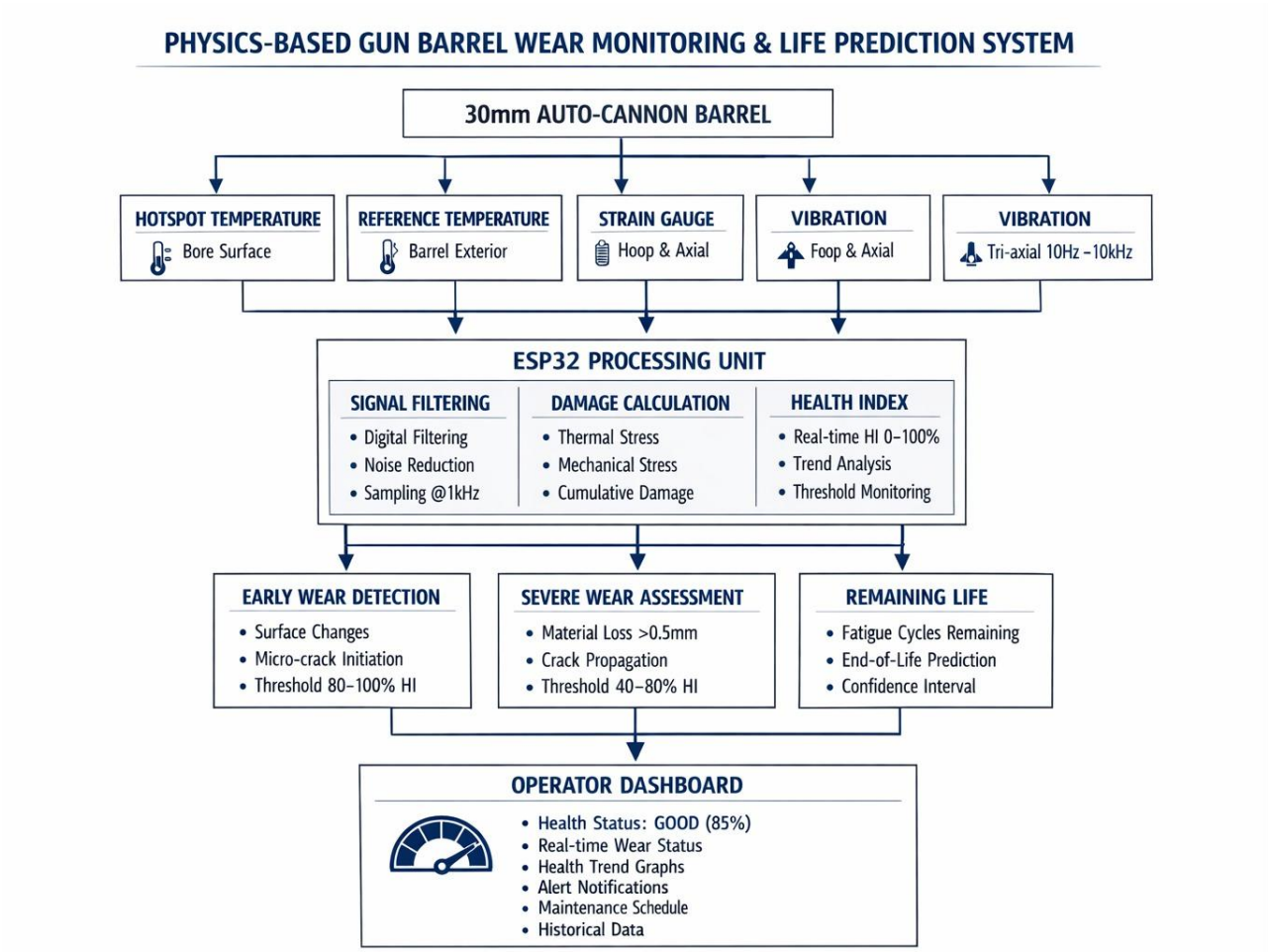
Another goal of the invention is to project cumulative fatigue damage based on well-known fatigue theories, which allows predicting the wear progress and stay safe firing cycles of the barrel.The other goal is to create a structural health index and condition classification, i.e., safe, warning, or critical, to help the operators and maintenance staff to make prompt decisions on maintenance.

Another goal of the invention is to minimize premature replacement and cost of barrels and maintenance since the invention is expected to allow predictive maintenance, i.e. based on real structural condition and not on some arbitrary firing-cycle limit.The other goal is to make weapon systems more operationally safe and reliable through the early detection of structural degradation including excessive stress, thermal damage, or fatigue.One of the last tasks of invention is to have a small, computationally effective monitoring architecture that can be incorporated into already existing defense weapon systems without having to

modify either the firing mechanism or the ammunition configuration.

Brief Description of the Drawings

The invention may be better understood with reference to the following illustrative components:



Detailed Description of the Invention

1. System Architecture

The Physics-Based Gun Barrel Wear Monitoring and Life Prediction System is a conceptualized framework of structural health monitoring that can be used to assess the thermo-mechanical health of a gun barrel during active firing strokes. Using the system architecture, several sensing components, embedded

processing unit, and analytical modeling algorithms, the constant observation of stress, deformation, and fatigue buildup in the barrel structure is performed.

The architecture starts with 30 mm auto-cannon barrel which is the main structural element under monitoring. When firing, the barrel undergoes severe internal pressure, sudden temperature variation and intermittent mechanical stress. These situations cause stress and material degradation which should be measured and analyzed to determine wear development and serviceable life left.
In order to measure these functional parameters, the system will have a number of sensor modules.
Thermocouples are installed at the bore part at high temperatures to determine the temperature of the hotspots and other temperature sensors are at the outer part of the barrel to record the temperature of the reference area. Through these measurements, the system can estimate thermal gradients and also measure thermal stress produced during firing. Moreover, to monitor strain the strain gauge is placed on the surface of the barrel to record the hoop strain and axial strain which is a result of internal firing pressure and structural expansion. The system also has a tri-axial vibration sensor that records dynamic vibration and shock signal when fired at the projectile.

The results of these sensors are sent to an internal processing unit which is usually an ESP32 microcontroller that does signal conditioning and data processing. It has a processing unit that involves digital signal filtering, noise reduction, and controlled sampling to guarantee good data acquisition despite high vibration working conditions.

The system carries out analytic computations based on physics after the process of preprocessing. Some of these calculations involve radial, hoop, and longitudinal stresses in equations of thick-cylinder pressure vessels, estimation of thermal stresses in the concepts of thermal expansion, and calculation of strain energy density on the barrel material. The system also uses cumulative fatigue

damage models (e.g. Miners rule) to determine the progressive structural damage as a result of repeated firing cycles.
According to the calculated parameters, the system creates a Health Index (HI) which shows the existing structural state of the barrel. The monitoring system also measures wear progress using steps like early wear detection, severe wear measurement, and remaining life estimation, which enable the operators to obtain the safe operating condition of the weapon system.

This processed information is then finally shown on an operator monitoring dashboard that gives real time health status, wear trends, maintenance alerts and remaining firing cycles predictions. Through this architecture, predictive maintenance is possible, operational safety is enhanced as well as efficient use of gun barrel service life.

2. Defect Classification Categories

- The Physics-Based Gun Barrel Wear Monitoring and Life Prediction System divides barrel degradation into verifiable defects depending on the calculated parameters of thermo-mechanical stress, strain, vibration response, and cumulative wear fatigue.
- These classifications enable operators and maintenance workers to know the extent to which structural degradation is taking place and take necessary preventive measures before severe failure takes place.

- The initial one is Early Wear Detection that is the first phase of structural degradation. This is where the slight surface wear, thermal expansion and slight differences in strain patterns can be observed. The system recognizes the initial signs like slight variations in the temperature distribution, small dislocation of the hoop strain, and the onset of the micro-crack formation in the material of the barrel.

- The early indications can be monitored to plan the maintenance even before the wear is serious.Moderate Wear or Progressive Damage is the second category and here the repetitive firing cycles result in significant fatigue build up and growing thermo-mechanical stress in the barrel structure.

- During this phase, the system can identify greater strain values, greater vibration levels, and greater thermal gradient between the bore surface and outer barrel surface. These criteria characterize gradual material overstrain and gradual growth of micro-cracks within the wall of the barrel.

- The third type is Severe Wear or Critical Damage that implies a high extent of structural degradation. The system could determine the presence of abnormal stress levels, strain deformation, crack propagation, and/or heavy material loss in the bore section at this point. Such conditions imply that the barrel is almost at the limit of the safe working life and it needs inspection or replacement.

- According to these classifications, the system creates a structural health status which may fall in the classification of Safe, Warning or Critical. The classification assists the operators to quickly gauge gun barrel condition and aids in predictive maintenance, which ensures operational safety, as well as optimizes use of barrel service life.

3. Operation of the System

During operation:

- Physics-Based Gun Barrel Wear Monitoring and Life Prediction System follows the structural state of the gun barrel during the firing spurts continuously. The temperature sensor of hotspot is mounted on the surface of the bore and reference temperature sensor is mounted on the periphery of the barrel to measure the temperature and external temperature respectively.

- Meanwhile, strain gauges provide measurements of hoop and axial strain created by internal firing pressure and thermal expansion and a tri-axial vibration sensor provides dynamic vibration measurements created during projectile discharge.

- The sensor outputs are sent to the embedded processing unit where signal conditioning steps are carried out to provide correct data acquisition at high-vibration conditions that include digital filtering, noise removal and controlled data sampling.

- The system follows with the processed sensor data to make physics-based analytical computations, such as the calculation of radial, hoop, and longitudinal stresses by equations of thick cylinder stresses. Dynamics of thermal stress that arise as a consequence of the fast-varying temperature value are also assessed.

- According to the estimated values of stress, the system uses fatigue models to compute strain energy density and cumulative fatigue damage. The calculations enable the system to predict wear ventures in the barrel framework and calculate the number of safe firing cycles left.

- The parameters calculated are then used to produce a Health Index (HI) which indicates the state of the structure of the barrel. The system can categorize the status of the barrel as SAFE, WARNING or CRITical depending on the measured wear level.

- The outcomes are presented on the operator monitoring dashboard where the maintenance staff can monitor the wear patterns, get notifications, and take necessary maintenance measures.By means of constant monitoring and real time analysis assessment, the system offers effective predictive maintenance capacity, enhances operational safety, and efficient use of gun

barrel service life.

4. Optimization Techniques

- The Physics-Based Gun Barrel Wear Monitoring and Life Prediction System uses a number of optimization methods to enhance the accuracy, efficiency, and reliability of the process of monitoring. The techniques will make sure that the system is capable of doing continuous structural health assessment with stable operation in high temperature and high-vibration conditions.Signal conditioning and filtering is one of the techniques of optimization in the system.

- The sensors will be used in a hostile firing environment, and therefore the data obtained will be noisy and subjected to distraction through vibration and shock. To enhance accuracy of measurement in the system, digital filtering and noise reduction methods are used prior to making analytical computation. This is to make sure that sensor signals that are processed properly reflect the physical conditions of the barrel.

- The second way of optimization is to have an efficient data sampling and processing in the embedded microcontroller. The system takes controlled sampling intervals to record the temperature, strain and vibration data without any superfluous computational load. This is a method that saves processing load and power and yet provides good real-time monitoring.

- The physics-based analytical models used to optimize performance also make the system optimized by balancing performance with the use of data-driven methods alone. The system can reliably estimate the wear by the use of known equations of mechanical engineering, i.e. the thick-cylinder stress analysis equation, thermal stress relations equation, and cumulative fatigue damage model with low level of computations.

- Moreover, the health index optimization on the basis of threshold is employed to categorize the state of the barrel as a safe, warning, or critical condition. These limits are established depending on materials properties, stress limits of operation, and fatigue life computations. This enables the monitoring system to detect very fast abnormal structural behavior and send out maintenance alerts when needed.

- Lastly, the system also has built in effective system architecture and sensor integration which means that the system will monitor effectively without disrupting the normal functioning of the weapon. The overall outcome of these optimization techniques is a good performance of the system, correct wear forecasting and efficient predictive maintenance of gun barrel systems in harsh defense conditions.

Advantages of the Invention

- The Physics-Based Gun Barrel Wear Monitoring and Life Prediction System offers a number of benefits that will lead to improvement of reliability, safety, and efficacy of gun barrel maintenance in military applications.

- Real-time structural observation of the gun barrel in the course of firing operations is one of the key benefits of the invention. Through constant monitoring of temperature, strain and vibration parameters, thermo-mechanical stress and structural degradation can be accurately monitored as it happens.

- The other notable benefit is that of predictive maintenance. The system does not wait until the barrel has completed a certain number of firing cycles before it is replaced or manually inspected, rather, it uses wear progression estimates to predict the number of safe firing cycles the barrel has left. This assists the maintenance staff to undertake maintenance in good time and

prevent unforeseen breakdowns.

- The invention also enhances the operational safety in that it identifies the early indication of damage like excessive thermal stress, abnormal strain pattern and fatigue build up. This will enable operators to implement corrective measures earlier than the barrel deteriorates to a critical state.

- The other pro is that of physics based analytical models like thick-cylinder stress analysis and fatigue damage assessment which give a clear and consistent wear estimation without the use of complex artificial intelligence algorithms.

- The system is also small and easy to deploy since it can be fitted to the existing weapon systems with no need to make any alterations to the firing mechanism or ammunition set-ups.

- Lastly, the invention contributes to the optimal use of the service life of gun barrels through less timely replacement and better planning of the maintenance. This results in an increase in resource efficiency, decreased maintenance fees, and increased efficiency in defence systems.

Claims:

- A gun barrel wear monitoring and life prediction system based on temperature sensing elements set up to measure bore hotspots temperature and outer barrel surface temperature during firing cycles, strain measurement sensors set up to measure hoop and axial strain generated by the internal firing pressure and thermal expansion, vibration sensing elements set up to detect dynamic vibration signals created in the event of discharging a projectile, an embedded processing unit set up to receive and process sensor data, and an analytical computation module set up to analyse thermo-mechanical stresses, fatigue loss and wear development of the gun.

- The system as per claim 1 which considers the thermal gradients on the bore

surface and the exterior barrel surface to determine thermal stress in firing operations. The system of claim 1, where the measurements of strain are based on the identification of deformation patterns that follow the changes in the radial expansion and longitudinal stress on the barrel structure. This system according to claim 1 where radial stress, hoop stress and longitudinal stress are calculated by the analytical computer module on the basis of thick 增 cylinder pressure vessel equations. The system as asserted in claim 1, where the embedded processing unit conducts signal conditioning such as filtering, noise rejection and controlled sampling in order to guarantee sound sensor data acquisition in high-extension conditions of high vibration.

- The system in assertion 1 in which cumulative fatigue damage is calculated by the fatigue life analysis models that calculate wear progression as a result of repeated firing experiences. The system in claim 1, where the system is used to create a structural Health Index (HI) that is an indicator of the state of the gun barrel reliant on the computed thermo-mechanical measures. Claim 1 The system described in the system claim and claim 1 whereby the structural health condition is categorized under the operation settings including Safe, Warning, and Critical by pre-determined levels of stress and fatigue measures. The system as stated in claim 1, where the monitoring results such as wear status, fatigue cycles to be consumed as well as structural health indicators are put on an operator monitoring interface.

- A wear and service life predictive control method of a gun barrel which consists of the following steps (a) continuously measure temperature, strain, and vibration sensor on the barrel, (b) process sensor data in an embedded processing unit, (c) calculate thermo-mechanical stress and fatigue life in physics based analytical models, (d) determine wear state and service life of the barrel (e) generate maintenance alerts based on the analysed structural health condition.

Independent Method Claim

- One of the ways to check the wear and determine the service life of a gun barrel, the process includes: measurements of the operations of the gun barrel continuously during firing with the help of temperature sensors, strain gauges, and vibration sensors attached to the structure of the barrel. The temperature, strain and vibration data is sensed and sent to an inbuilt processing unit where the processing unit conditions the signal and preprocesses it and carries out filtering, noise removal and controlled sampling to form reliable operational data at high-vibration firing conditions.

- The technique also involves the calculation of thermo-mechanical stress parameters such as radial stress, hoop stress, longitudinal stress using thick-cylinder equations of pressure vessels on the basis of processed sensor output. The thermal expansion relations are used to determine the thermal stress developed as a result of quick temperature change in the barrel. The technique also includes the determination of strain energy density and cumulative damage of the material due to fatigue by use of fatigue life evaluation models to determine the wear progression in repeated firing cycles.

- According to the measured parameters, a structural health index that reflects the health of the gun barrel is produced. The technique also involves categorizing the barrel state as operational states such as Safe, Warning, and Critical, and therefore allows to predict the rest of the safe firing cycles and predictive maintenance of the weapon system.

Dependent Claims

- The procedure in claim 2 whereby the temperature sensors are placed at the bore surface and external barrel surface, to determine thermal gradients created during firing performances.The approach in the alleged claim 2 where strain gauges are set to the value of hoop strain and axial strain

generated as a result of internal pressure and thermal expansion of the barrel.

- The procedure as presented in claim 2 which involves vibration sensors that pick vibration signals that are dynamic during projectile release to determine mechanical stress conditions.The approach described in claim 2, whereby the internal processing unit is used to execute signal conditioning, having digital filtering and noise elimination and controlled sampling of the data in order to enhance measurement accuracy.

- The procedure as specified in claim 2, wherein the thermo-mechanical stress such as radial stress, hoop stress and longitudinal stress are determined by thick cylinder stress equations.The process as stated in claim 2, in which the cumulative fatigue damage of the barrel material is the estimation of fatigue life assessment models to ascertain wear progression.

- The procedure according to claim 2, where the system will produce a structural Health Index (HI) indicating the state of the gun barrel due to evaluated parameters of stress and fatigue.The approach in claim 2 in which the health index is applied to categorize the state of the operation of the barrel as Safe, Warning, and Critical to support the maintenance decision.The approach as described in claim 2, where the monitoring data in the form of the percentage of wear, fatigue cycles left, and the state of structure are represented as operator monitoring interface.

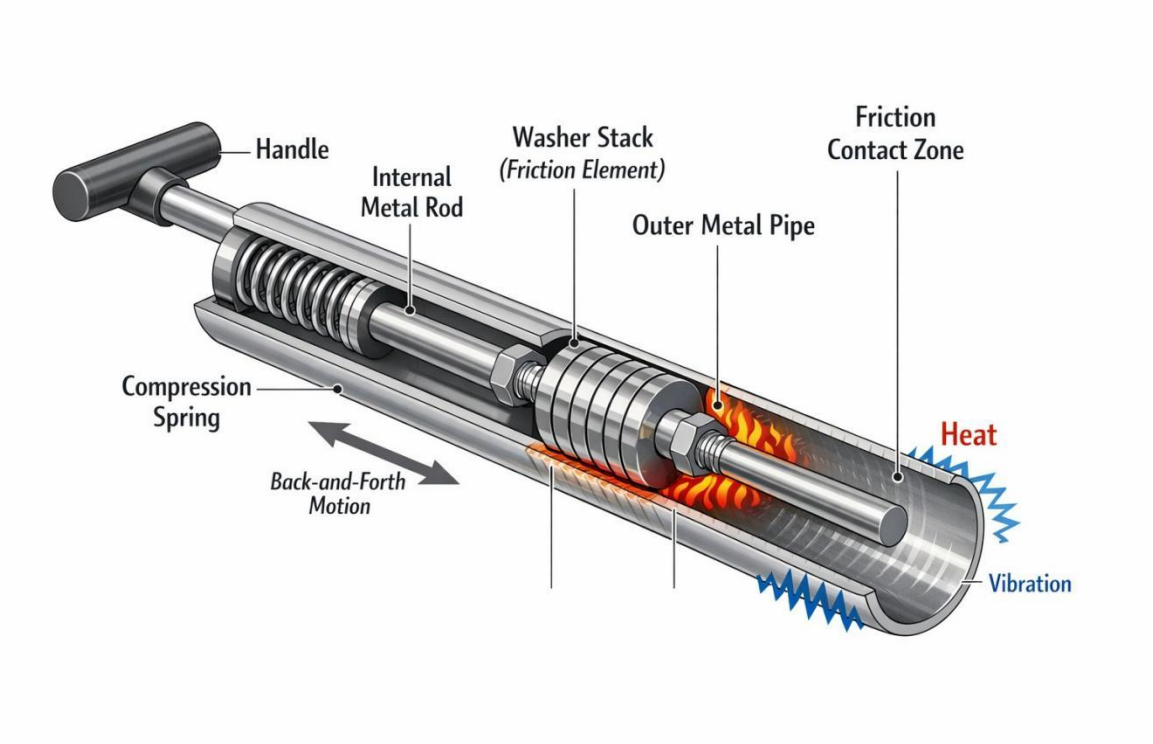
ABSTRACT

The current invention is associated with Physics-Based Gun Barrel Wear Monitoring and Life Prediction System that aims to assess structural damage of gun barrels used in high-pressure firing which is a cyclic exercise of medium caliber guns. Gun barrels undergo severe thermo-mechanical loads such as high internal pressure, elevated temperature increase, and shock loading during firing operation. These circumstances eventually cause material fatigue, thermal stress build-up, micro-crack development and bore surface erosion that can decrease the structural integrity and working life of the barrel unless properly monitored.

The proposed system offers real time monitoring and predictive maintenance system by incorporating temperature sensors, strain gauges and vibration sensors in order to measure operational parameters of the gun barrel during the firing cycles. The sensor data received is interpreted in an embedded processing unit which makes signal conditioning and analytical computation processes relying on well known rules of mechanical engineering.

This system uses the equations of thick-cylinder stress, and the stresses generated by internal firing pressure are the radial, hoop, longitudinal stresses. The thermal stress due to quick change in temperature is measured based on thermal expansion relations and cumulative fatigue damage is measured based on fatigue life analysis models. Referring to these calculations, the system defines the wear state of the barrel and foretells the rest of the safe firing cycles.
A Health Index (HI) is made to indicate the structural status of the gun barrel and the system categorizes operational state either as safe, warning or critical. The findings are presented in a format of an operator monitoring interface where the maintenance staff can monitor wear trends and carry out preventive maintenance on time.

Prototype: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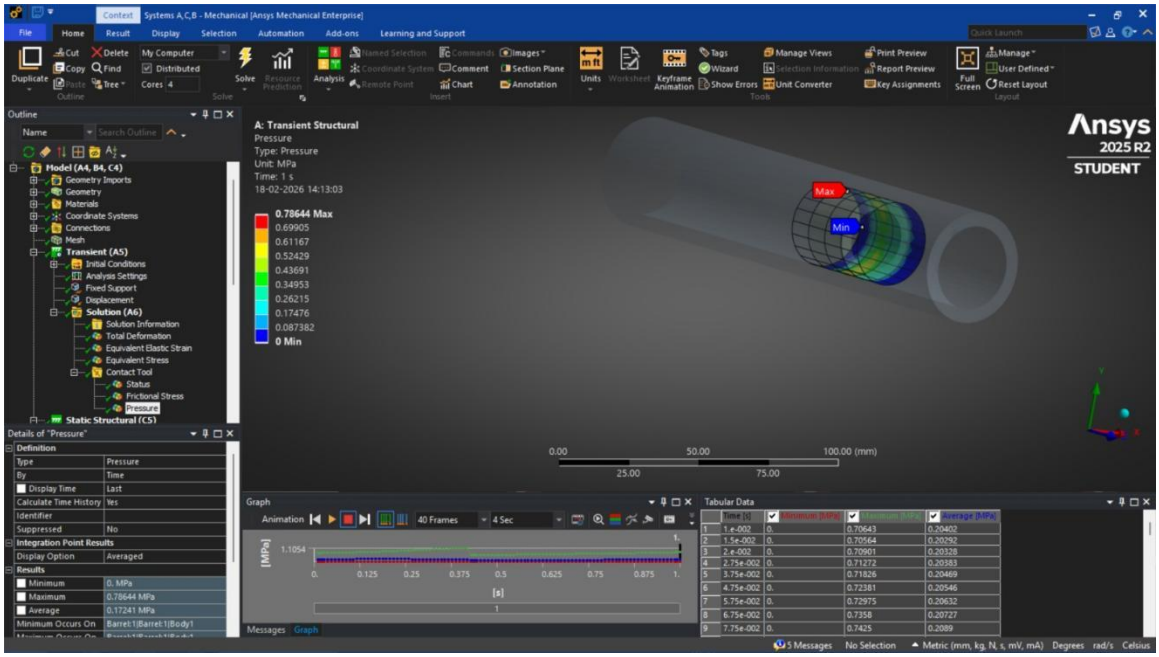

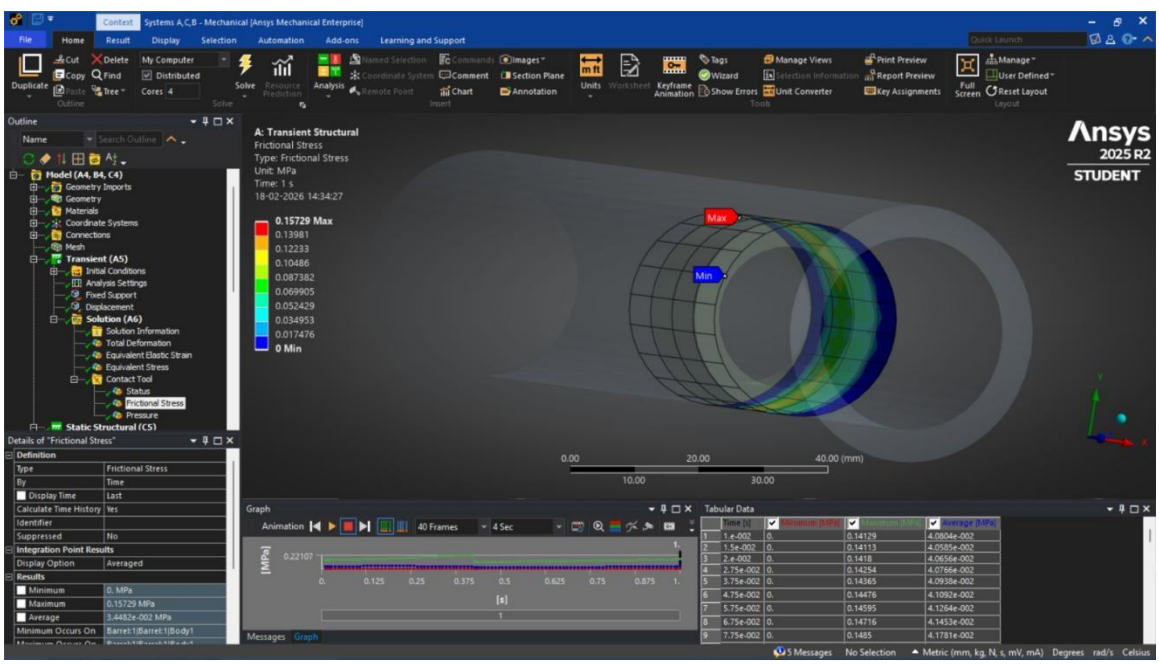
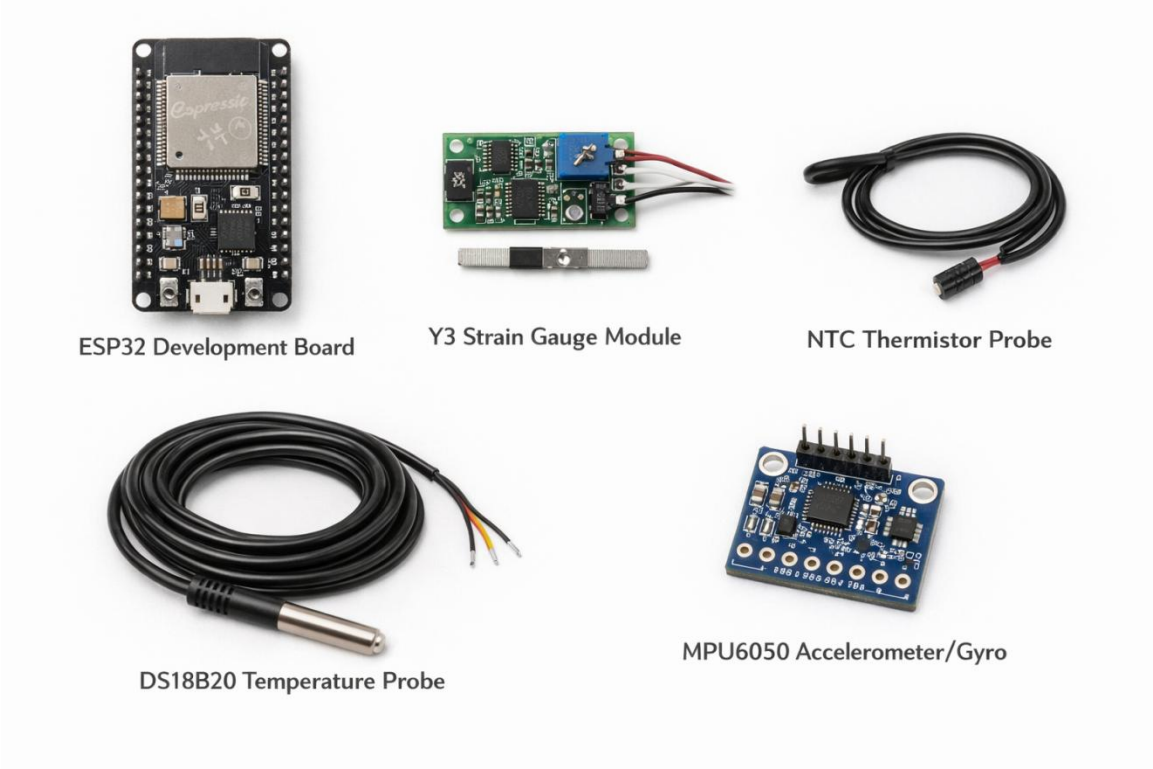
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